UNIVERSITY OF CALIFORNIA, BERKELEY
DEPARTMENT OF ECONOMICS

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RESEARCH AND TEACHING FIELDS:

PRIMARY
Microeconomic Theory

SECONDARY
Fair Division
Mathematical Economics

EDUCATION	**DEGREE**	**DATE**
UC Berkeley	Ph.D. in Economics	2027 (expected)
Thesis: Essays in Fair Division		
Committee: Federico Echenique, Chris Shannon, Quitze Valenzuela-Stookey, Yuichiro Kamada		
Kyiv School of Economics	M.A. in Economic Analysis	2020
Taras Shevchenko National University of Kyiv	B.A. in International Economic Relations	2018

WORKING PAPERS:

- The Power of Sperner's Lemma in Fair Division (Job Market Paper)
  I establish the existence of allocations that are simultaneously envy-free and weakly efficient in a broad class of allocation problems. I develop a unified existence proof that encompasses classical settings with divisible and indivisible goods while extending fair-division analysis to infinite-dimensional environments, including dynamic and state-contingent allocation problems. A quantitative adaptation of the proof yields a new algorithm for computing allocations that satisfy both envy-freeness and efficiency to an arbitrary degree of precision, applicable to a broad class of allocation problems.
- Multi-period Fair Division
  with Federico Echenique and Albin Erlanson
  We prove existence of envy-free and Pareto efficient allocation in a multi-period indivisible goods assignment problem. Since Birkoff-von-Neumann fails in this environment, the allocation problem cannot be reduced to individual-level consumption problem where total demand cannot exceed supply. Therefore, we must treat economy-wide allocations as a public good. We provide a market-based implementation by showing that such an allocation can be supported as a Lindahl equilibrium. We then look at a large economy and show that for large enough economies, the incentives to misrepresent agents' preferences vanish.
- Implementation in the Cake-Cutting Problem
  I establish an equivalence between lotteries over deterministic cake partitions and fractional cake allocations—partitions of unity. The former is a common tool for convexifying a set of deterministic allocations, while the latter is an insightful embedding used in mathematical fair division. By proving that these two objects are utility-equivalent, I provide an implementation foundation for a mathematical representation commonly used in fair division. In addition, I show that every simple partition of unity belongs to the convex hull of degenerate partitions of unity, which means it is implementable by a lottery with finite support.

PROFESSIONAL EXPERIENCE:

RESEARCH:
Consultant, Global Sovereign Debt & Currencies, Barings LLC (2020)
Lead Economist, National Bank of Ukraine, Department of Monetary Policy (2019)

TEACHING:
Teaching Assistant, Department of Economics, U.C. Berkeley (Fall 2021- Fall 2026)
Intermediate Microeconomics, Graduate Game Theory, Graduate Mathematical Tools for Economists.

FELLOWSHIPS AND AWARDS:

2026-2027	Doctoral Completion Fellowship, UC Berkeley
2020-2025	Departmental Fellowship, UC Berkeley, Department of Economics
2018-2020	FUIB Merit Fellowship, Kyiv School of Economics

OTHER INFORMATION:

Languages:	English, Ukrainian, basic German
Citizenship:	Ukraine

REFERENCES:

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